LIN7076 – Foundations of Computational Linguistics

Automata and context-free grammars

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Queen Mary University of London

Plan for today

Can we come up with simple rules summarizing a language or a fragment thereof?

- Last week, we discussed a fundamental limitation of regexes: they **cannot keep track of arbitrary occurrence counts**.
- We have also seen that regexes **cannot account for the internal structure of strings**, and the meaning differences that may arise from structural ambiguities.
- Today we'll discuss a graphical representation of regexes, **automata**, and show that they run into similar issues.
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Automata

Graphical representation of regexes: automata

- Regexes can be represented using diagrams called **automata**. The two formalisms are equally expressive.
- Just like regexes, automata express concatenation, union, and the Kleene star operation and can become increasingly complex by combining these atomic operations together.
- Automata have **states**, represented with **circles**. There's one **initial** state (where you start), and at least one **accepting** state (where you want to end).

Initial state

Accepting state

- States are connected with arrows labeled with words of the alphabet Σ (including the empty word ϵ).
- An automata **matches a sentence** if there is a path from the initial state to an accepting state whose successive arrows match the words of the sentence.

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- The following automaton accepts only one sentence, *the boy*. It corresponds to the concatenated regex the boy.



- More generally, regex **concatenation** is represented by **arrows between states**.

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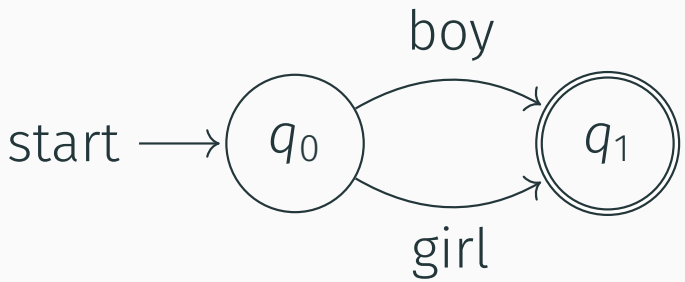
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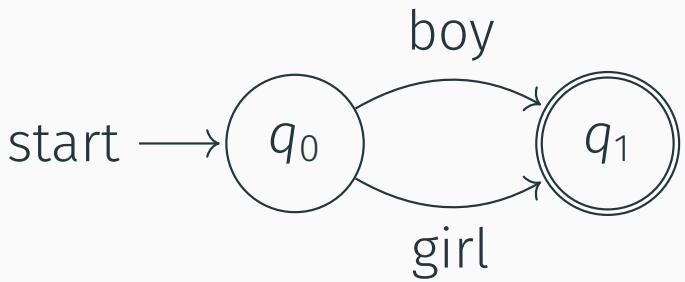
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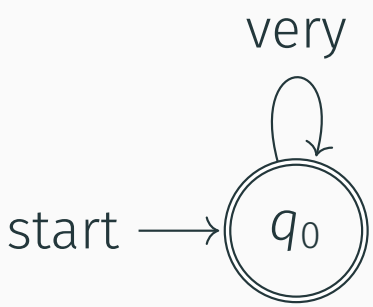
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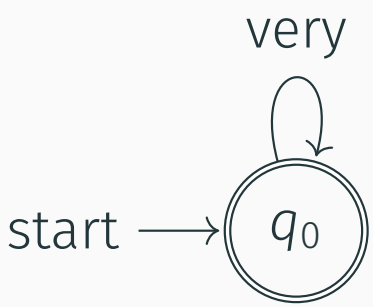
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Practice: automaton for intensified adjectival phrases

- Recall that our pattern for *the/a boy/girl*, *the/a tall boy/girl*, *the/a very tall boy/girl*... was: (a|the)((very*)tall|ϵ)(boy|girl).

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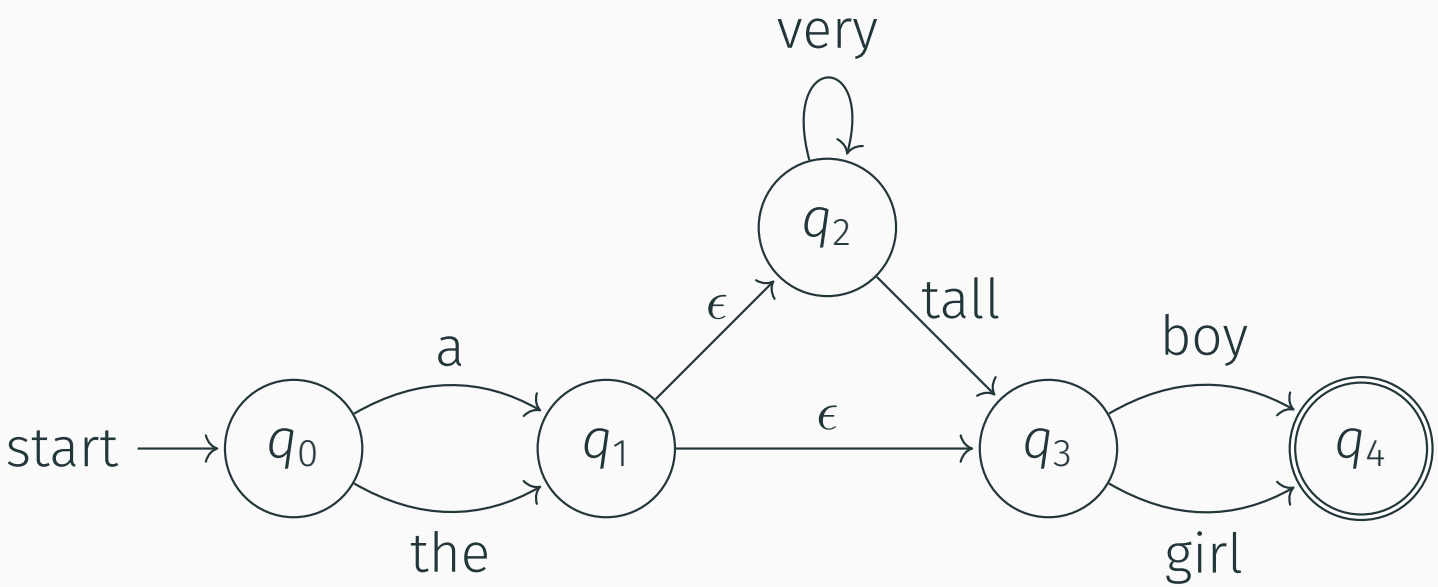
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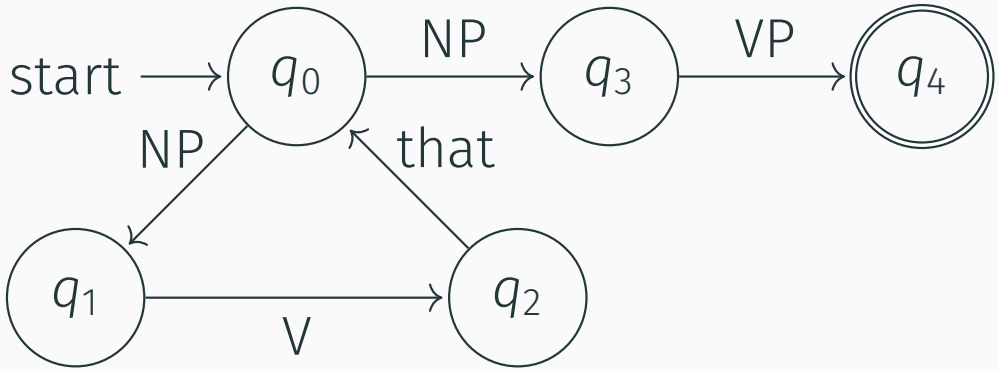
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Some extra challenges

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- Last week, we saw that regexes struggle with center-embedded structures, and ambiguous structures like *tall boys and girls*. **These challenges surface again with the equivalent automata**.
- Here is another problematic example.

(1) The photographer saw the bird on the alligator with long legs.

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Different readings of PPs: case 1

(1) The photographer saw the bird on the alligator with long legs.

- (1) can mean the action of seeing the bird was done on the alligator (i.e. the photographer is on the alligator), and the alligator has long legs.
- In that case, we say the alligator-PP modifies the verb (attaches "high"), and the long-leg-PP modifies the alligator (nested attachment).

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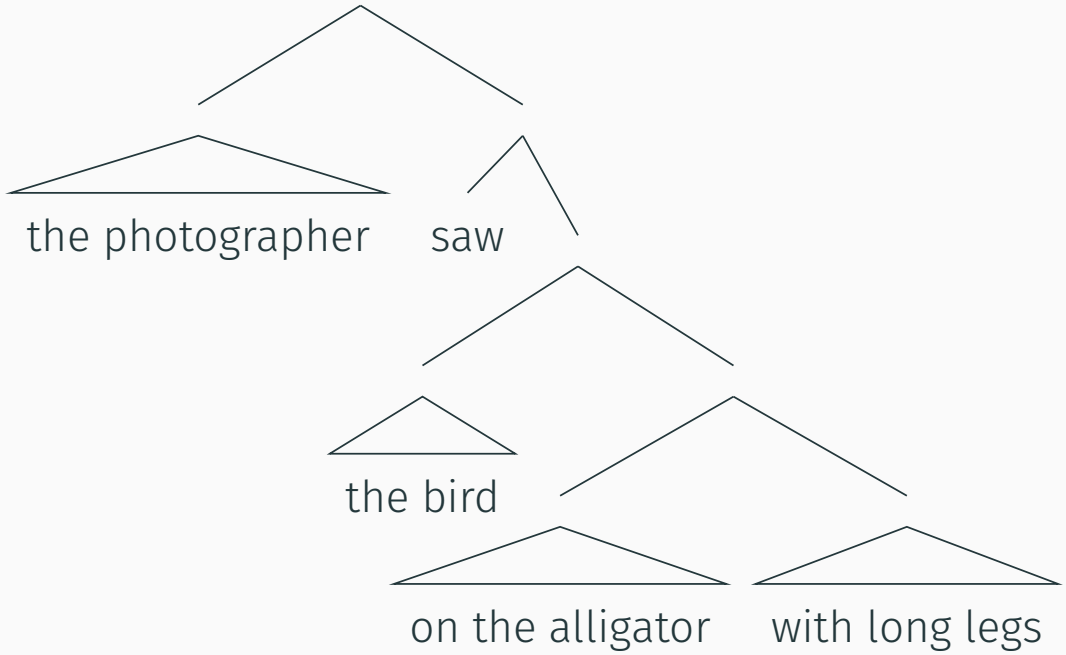
(1) The photographer saw the bird on the alligator with long legs.

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Different readings of PPs: case 4 (for completeness)

- The following reading is spelled out for completeness. The better see it, let's replace *with long legs* with *with binoculars*. This affects the plausibility of the sentence's various readings, but not really its core structure and dependencies.

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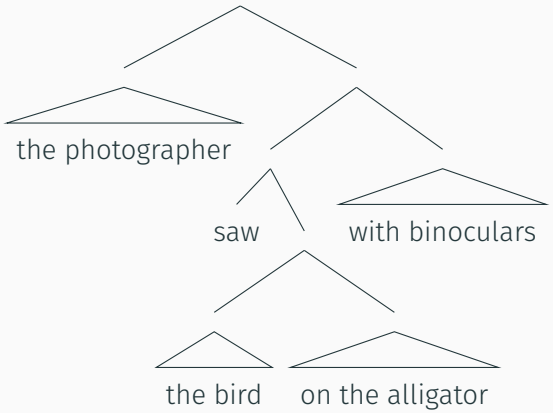
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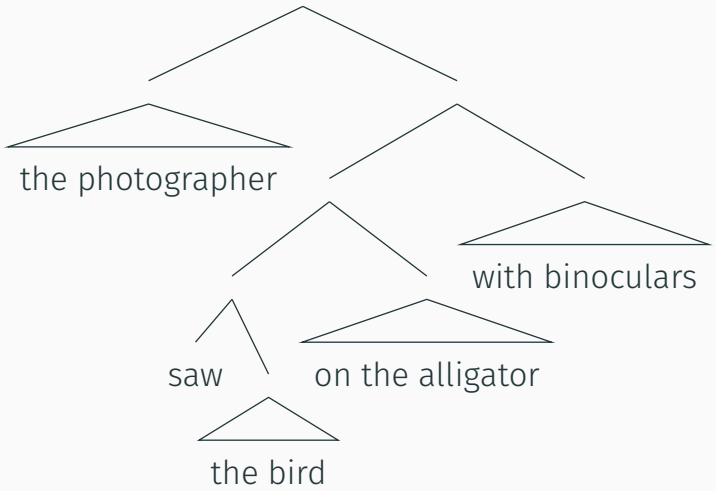
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Different readings of PPs: the impossible reading

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- A reading that is **impossible** to get is one where the photographer is on the alligator (i.e. the alligator-PP modifies the verb), and the bird has long legs (i.e. the long-leg-PP modifies the bird).
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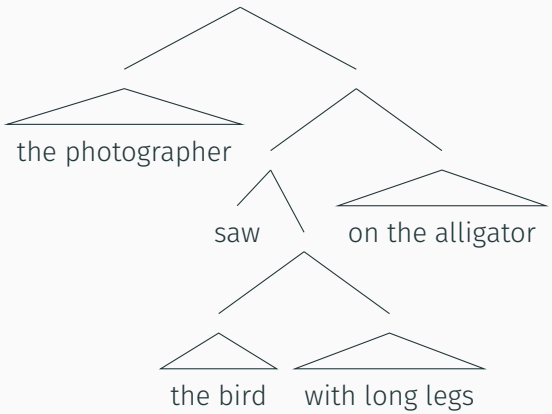
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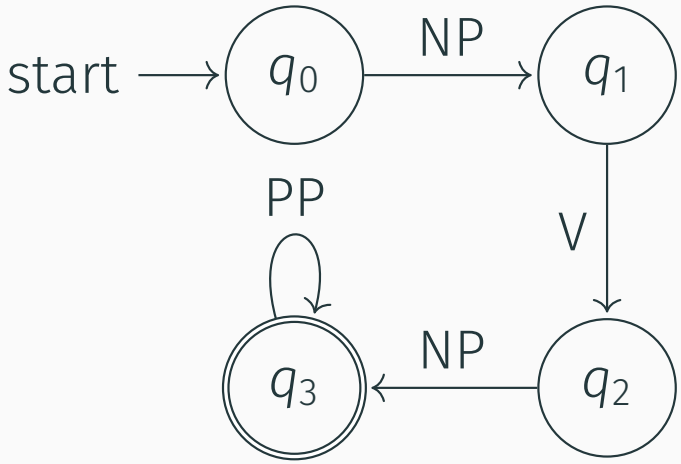
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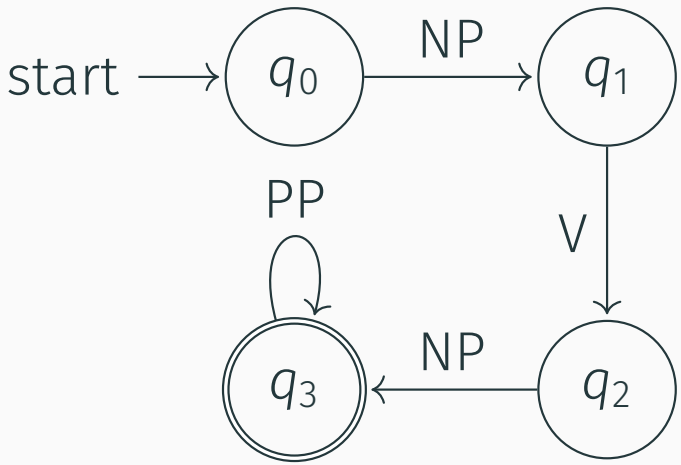
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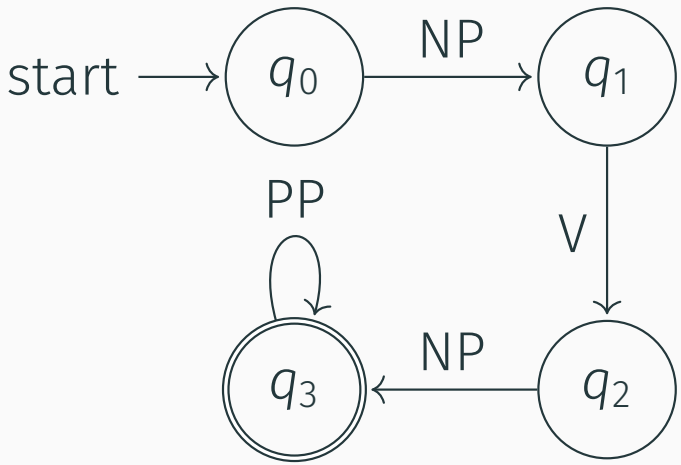


The challenge of PP-attachment ambiguities



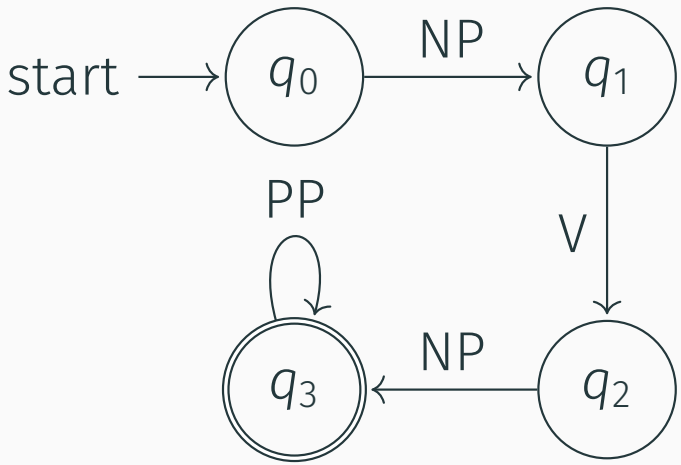
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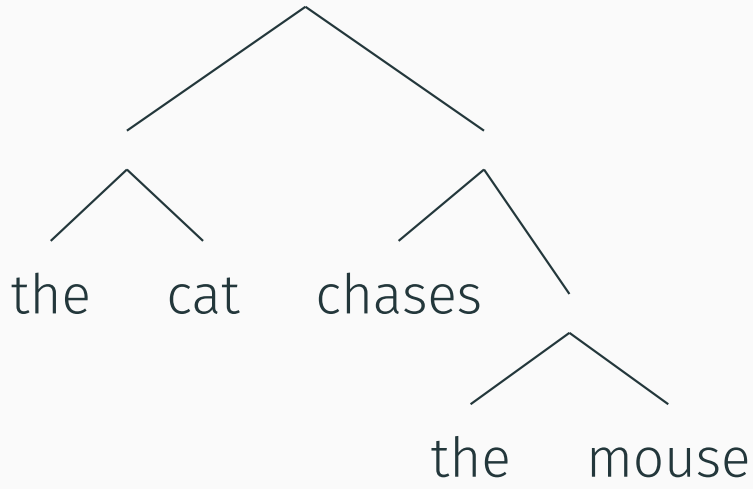


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Context-free grammars

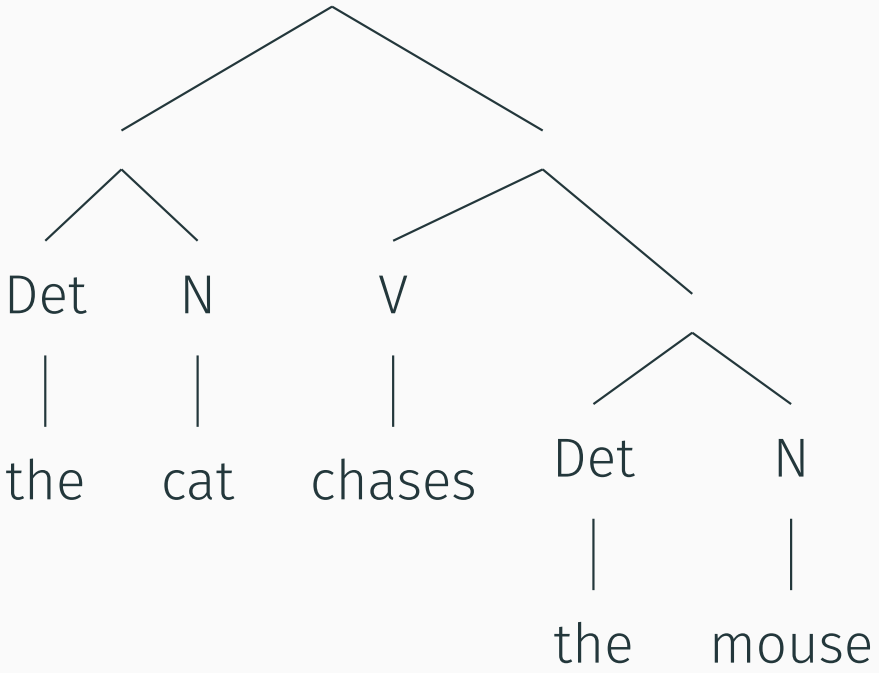
Intuitive Syntax trees

- The PP-attachment facts suggest **sentences adopt a hierarchical, nested structure**, represented as a tree.



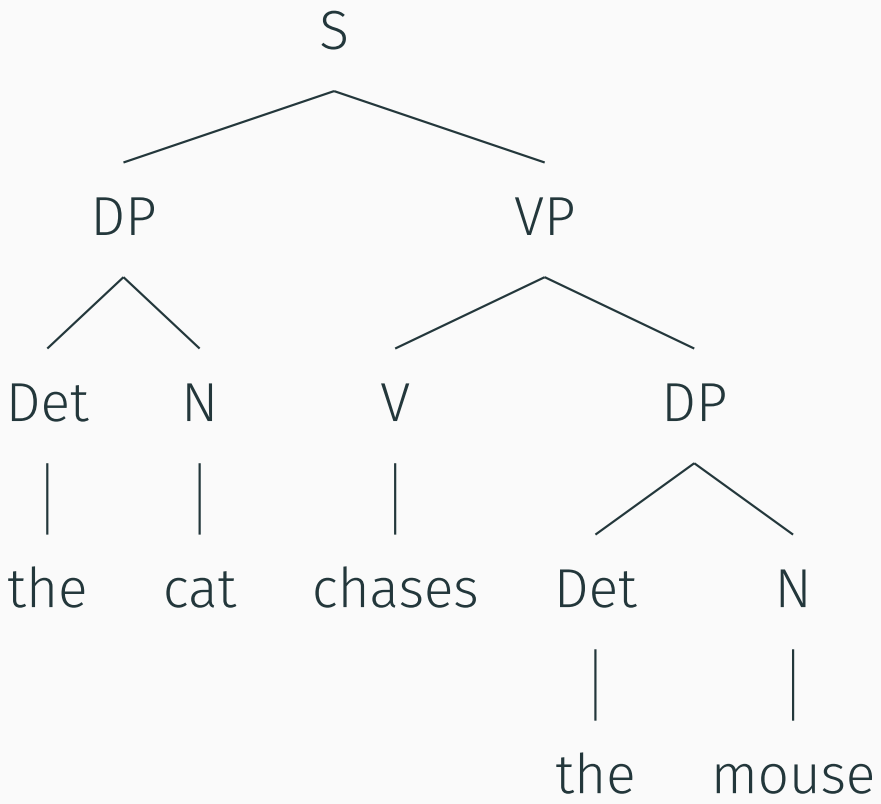
- Intuitive trees can be obtained by iteratively merging nodes together based on intuitions about what forms a unit (**constituent**).

Intuitive Syntax trees



- We now add Part-of-Speech labels to the lexical items.

Intuitive Syntax trees



- We also add labels to intermediate nodes according to how they "behave": DP=Determiner Phrase; VP=Verb Phrase, S=Sentence, NP=Noun Phrase, PP=Preposition Phrase...

Generating syntax trees

- Context-free grammars a formalisms generating syntax trees. They are defined as:
  - A finite set Σ of **terminals**, (leaves, i.e. actual words of the language).
  - A finite set V of **nonterminals** (intermediate nodes, i.e. phrase or category labels). V contains a special nonterminal S used to represent entire sentences (root).
  - A set R of **production rules**, i.e. input $\rightarrow$ output pairs, where inputs are single nonterminals (understood as mother nodes) and outputs are (non)terminals or concatenations thereof (understood as children node(s)). For instance,

corresponds to the production rule S $\rightarrow$ NP VP
- Pāṇini, a scholar in ancient India, came up with a grammar of Sanskrit following a very similar formalism in the mid-1st millennium BCE.

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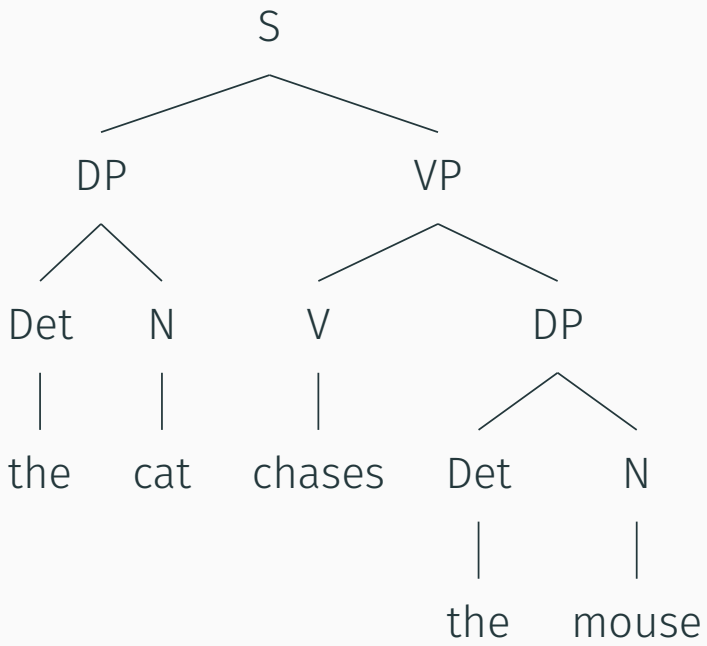
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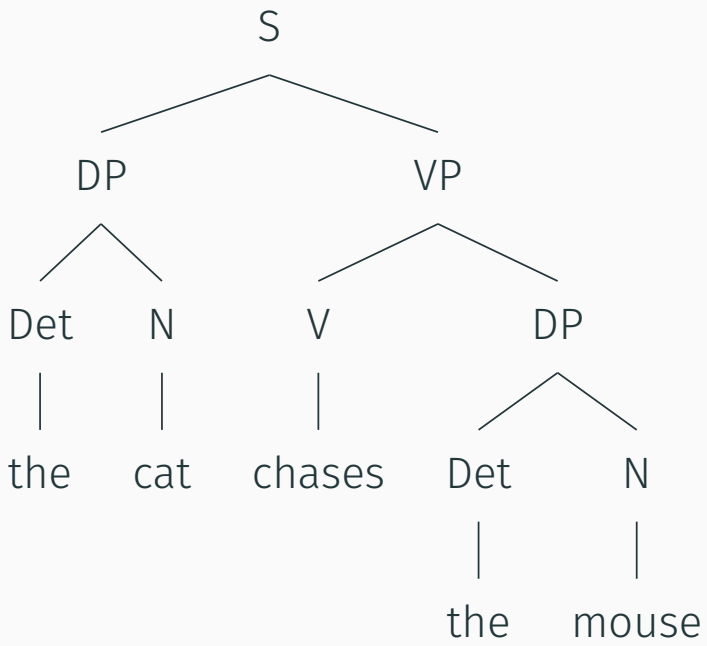
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Production rules practice



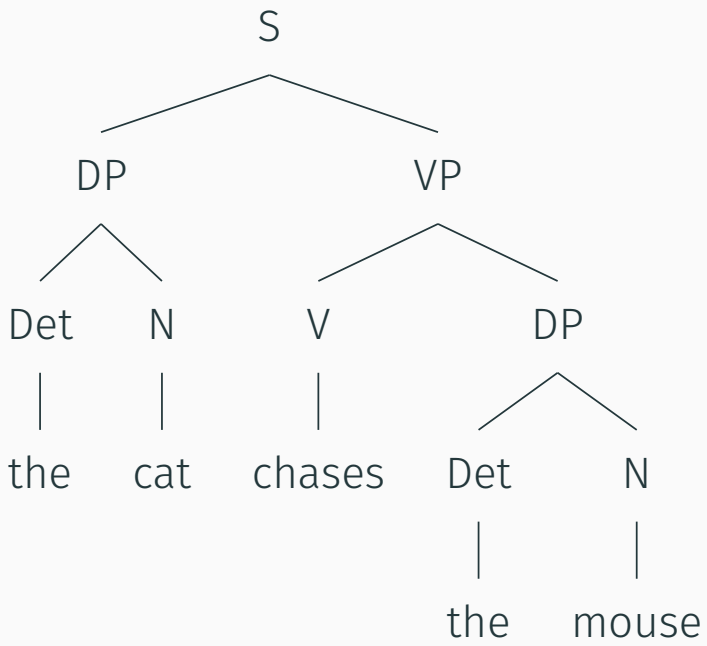
- What are the terminals?

Production rules practice



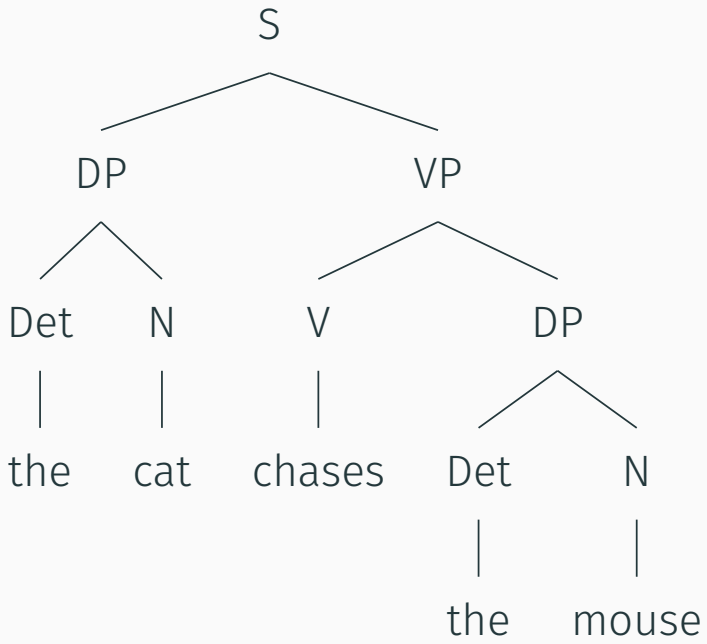
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Production rules practice



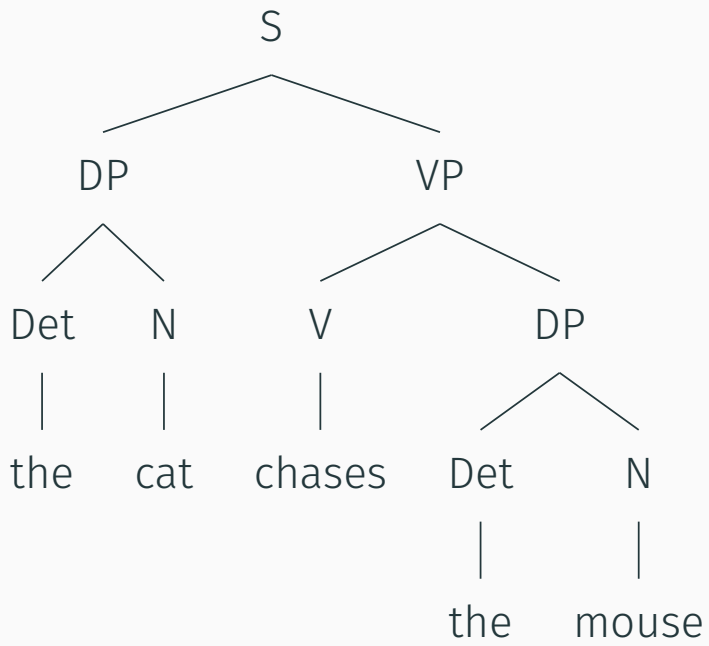
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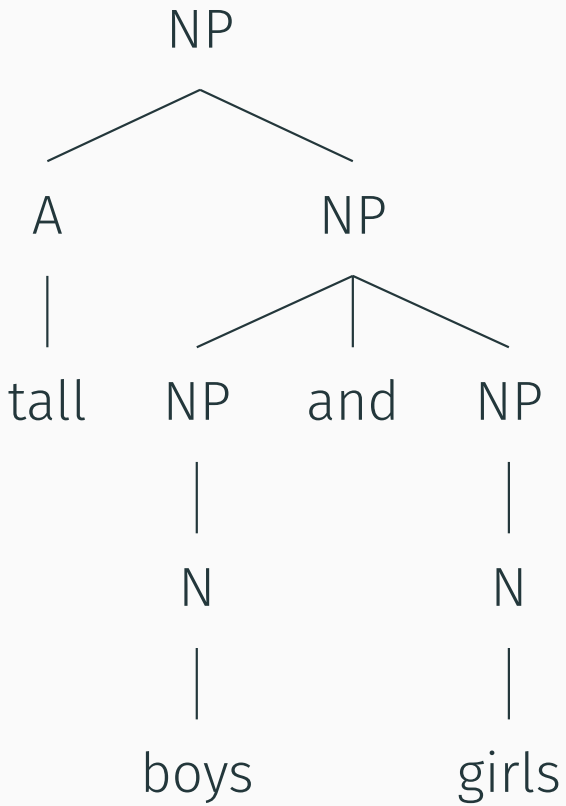
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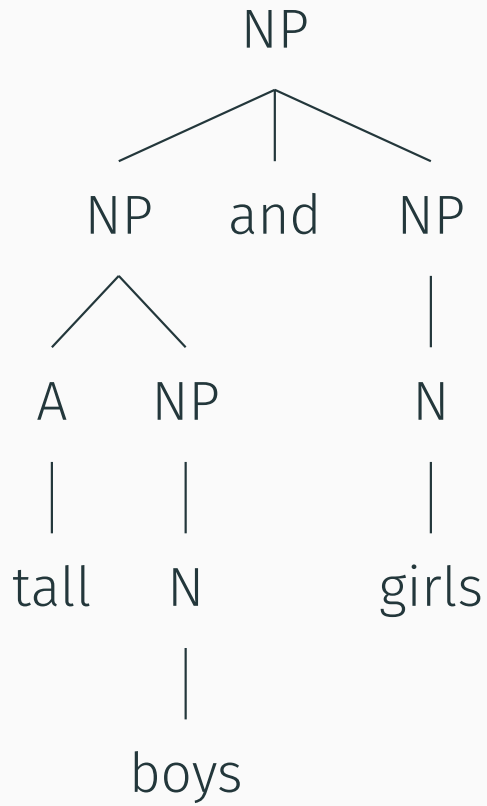


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Structural ambiguity 1: adjective attachment

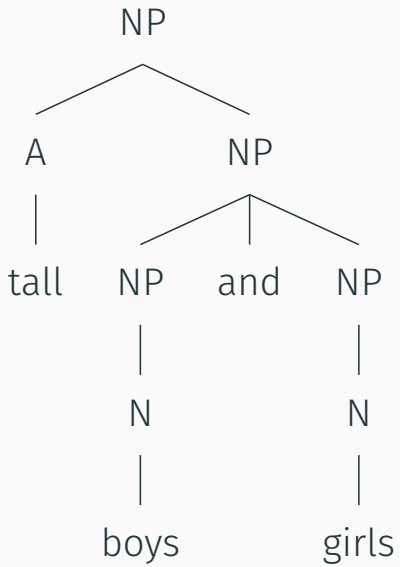
- Intuitively, which tree corresponds to which interpretation?

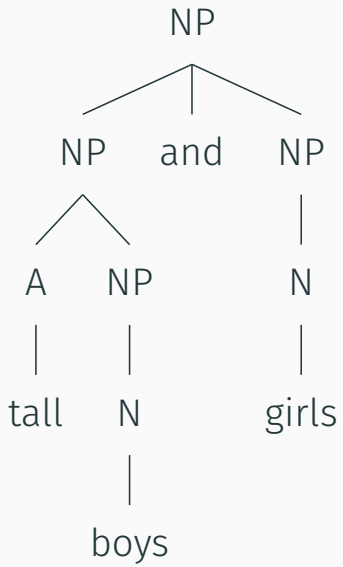




- Terminals? Nonterminals? Production rules?

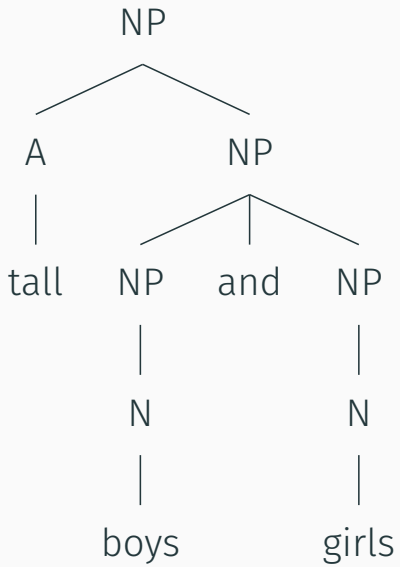
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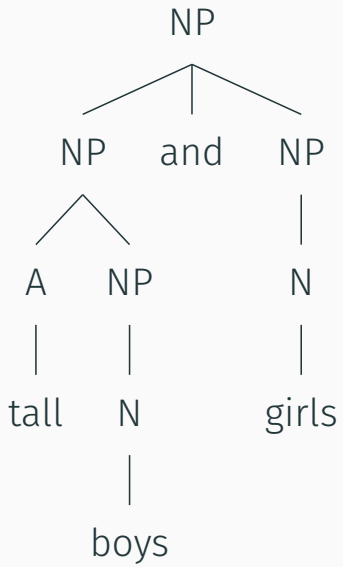




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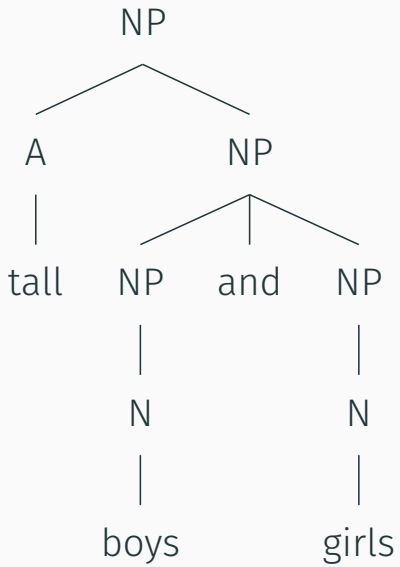
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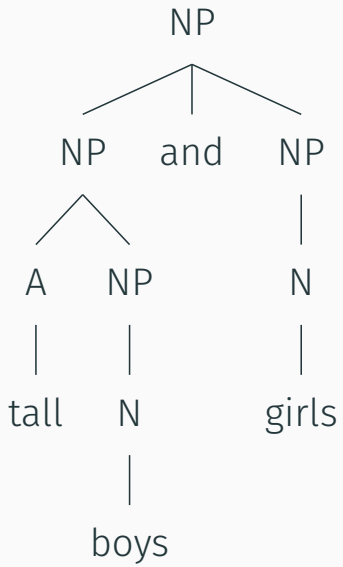




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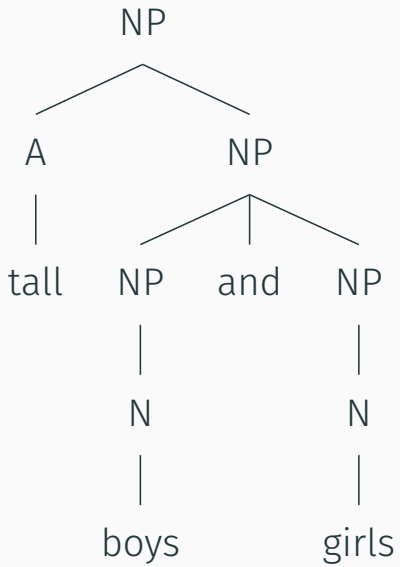
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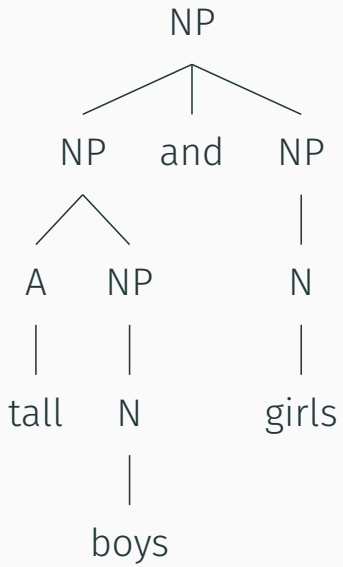




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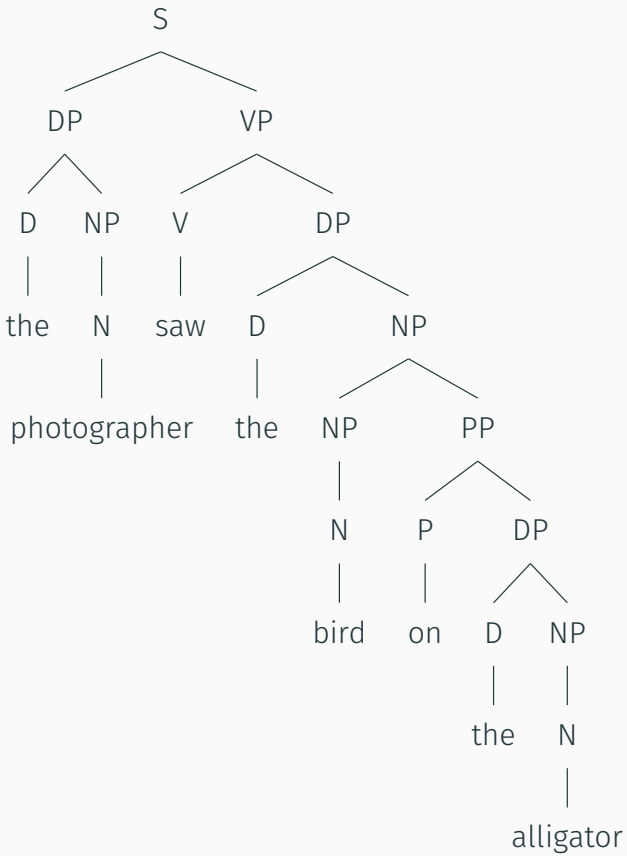


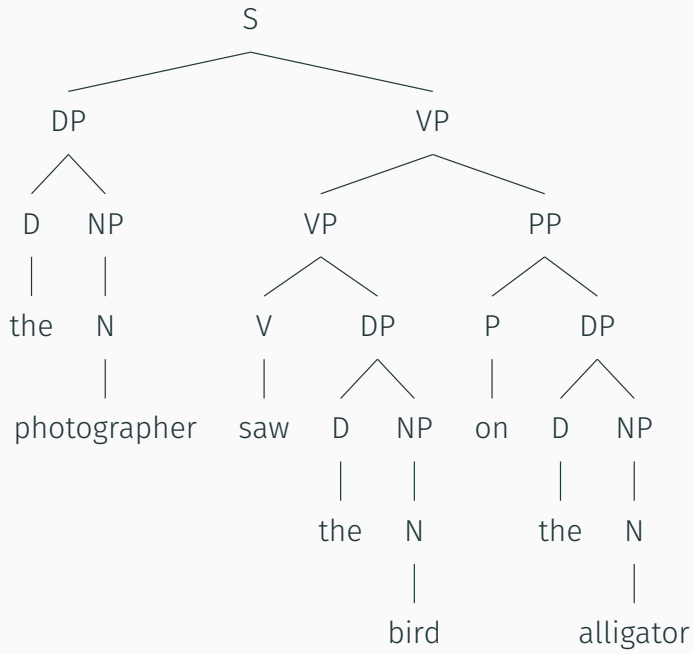


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Structural ambiguity 2: PP-attachment

(1) The photographer saw the bird on the alligator.





- Which tree intuitively yields which reading?

Structural ambiguity 2: PP-attachment

- Consider the grammar below. Which reading(s) does it capture?

S $\rightarrow$ DP VP
DP $\rightarrow$ D NP
VP $\rightarrow$ V DP
NP $\rightarrow$ NP PP
NP $\rightarrow$ N

PP $\rightarrow$ P DP
D $\rightarrow$ the
N $\rightarrow$ photographer, bird, alligator
V $\rightarrow$ saw
P $\rightarrow$ on

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- The only rule in which PP is an output (daughter node) is NP $\rightarrow$ NP PP. In that rule, PP's sister/mother node is NP, which means PP attaches to an NP. It models the *bird-on-alligator*-reading.

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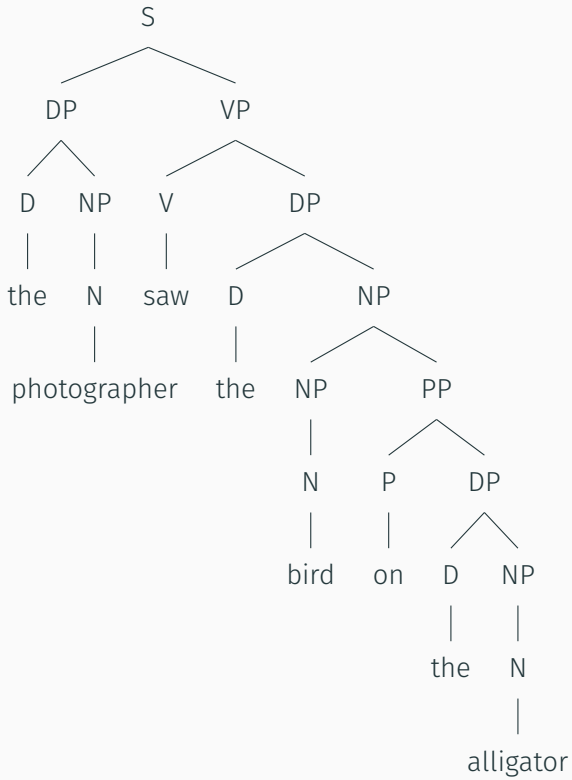
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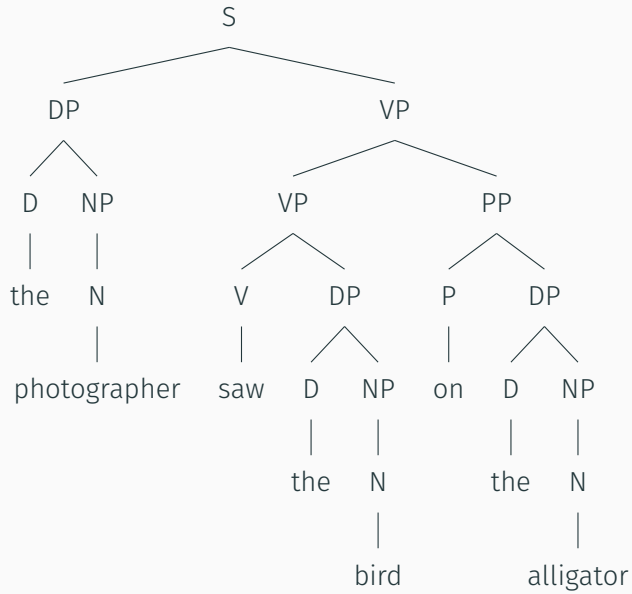
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Structural ambiguity 2': double PP-attachment

(1) The photographer saw the bird on the alligator with long legs.



Bird-on-alligator tree

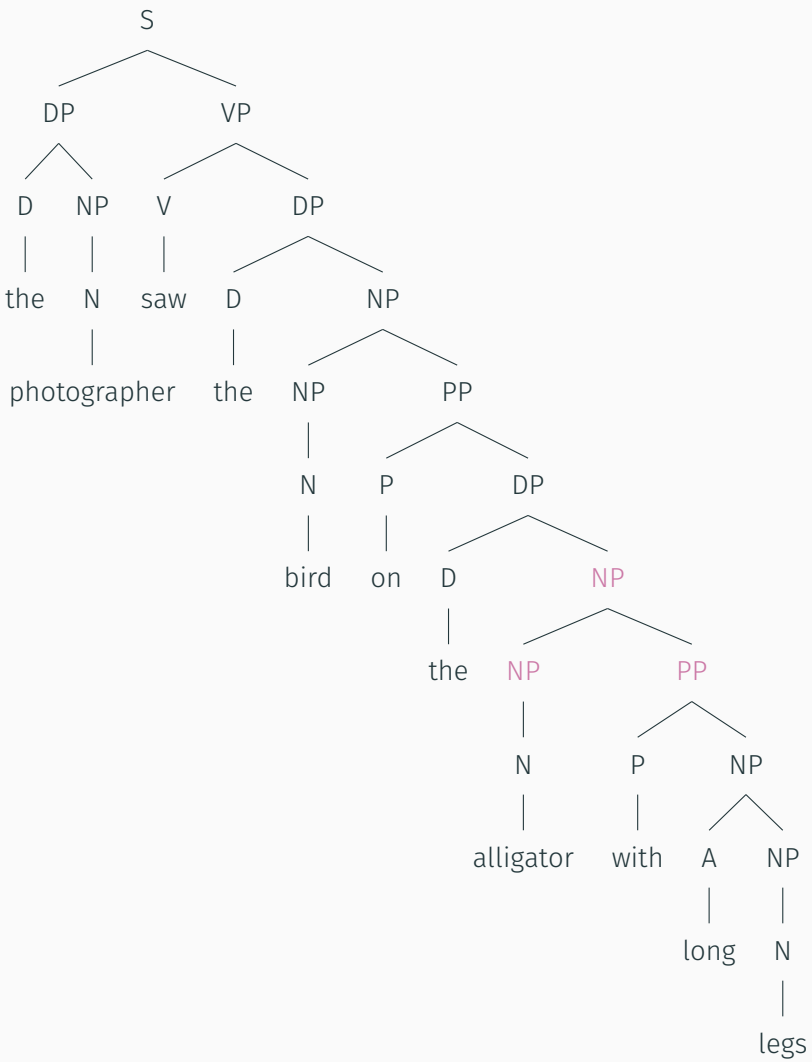


Photographer-on-alligator tree

- Two possible PP-introducing rules: VP $\rightarrow$ VP PP and NP $\rightarrow$ NP PP.
- Where can we "insert" the PP *with long legs* in the above trees? Which attachment site yields which reading?

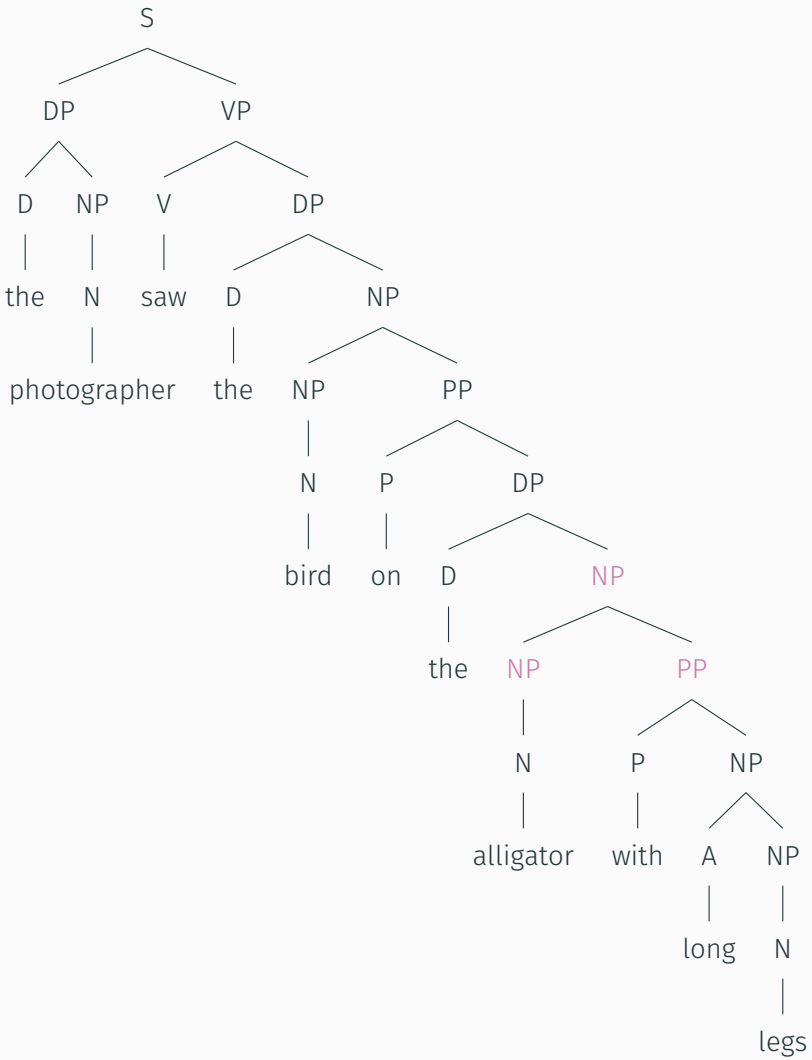
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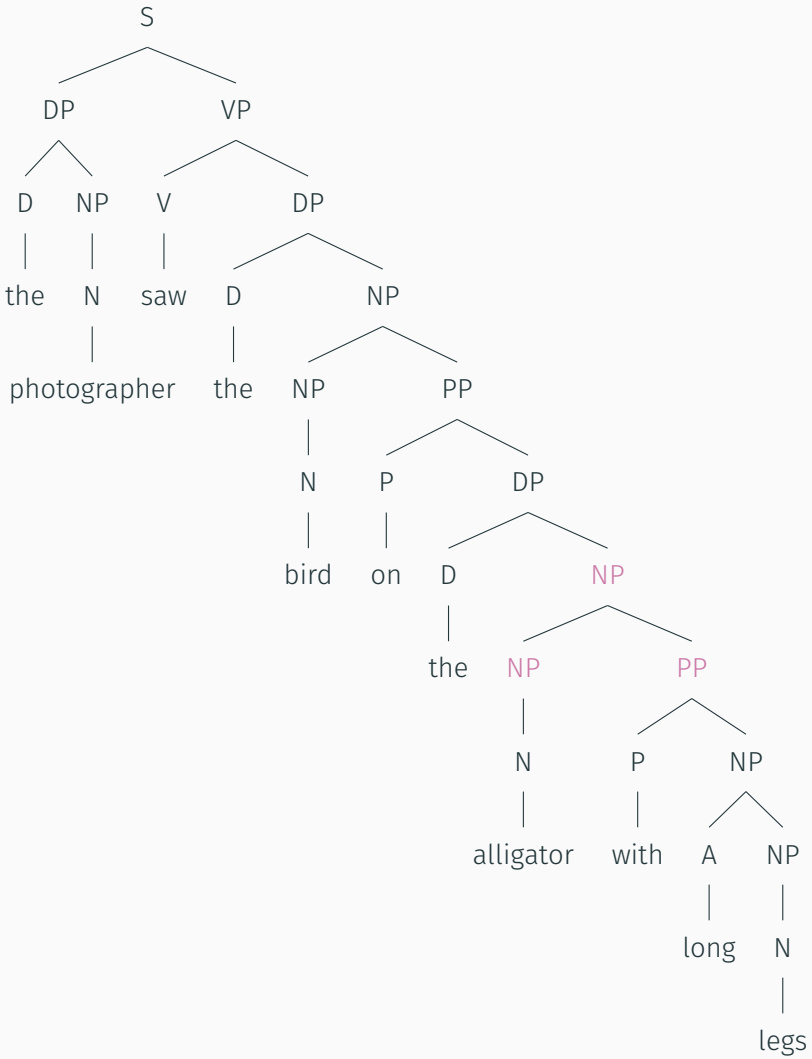
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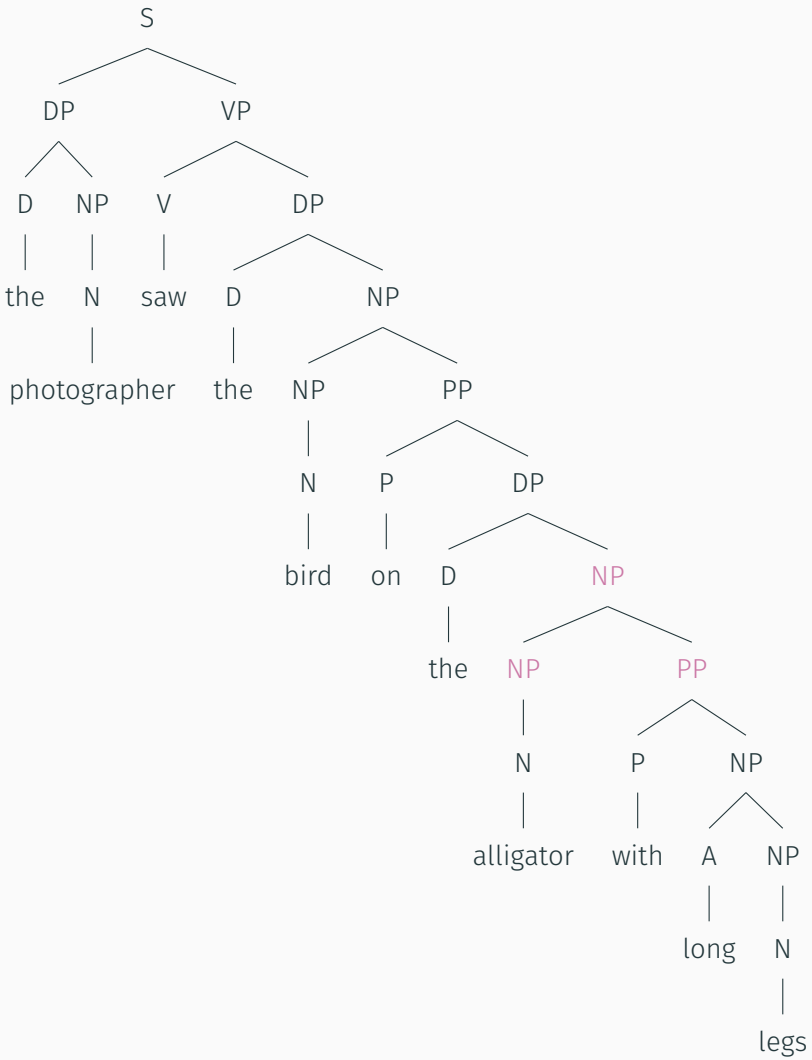
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- We add a PP $\rightarrow$ P NP rule (to get *with legs*), and a NP $\rightarrow$ rule (to get *long legs)*.
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Structural ambiguity 2': double PP-attachment

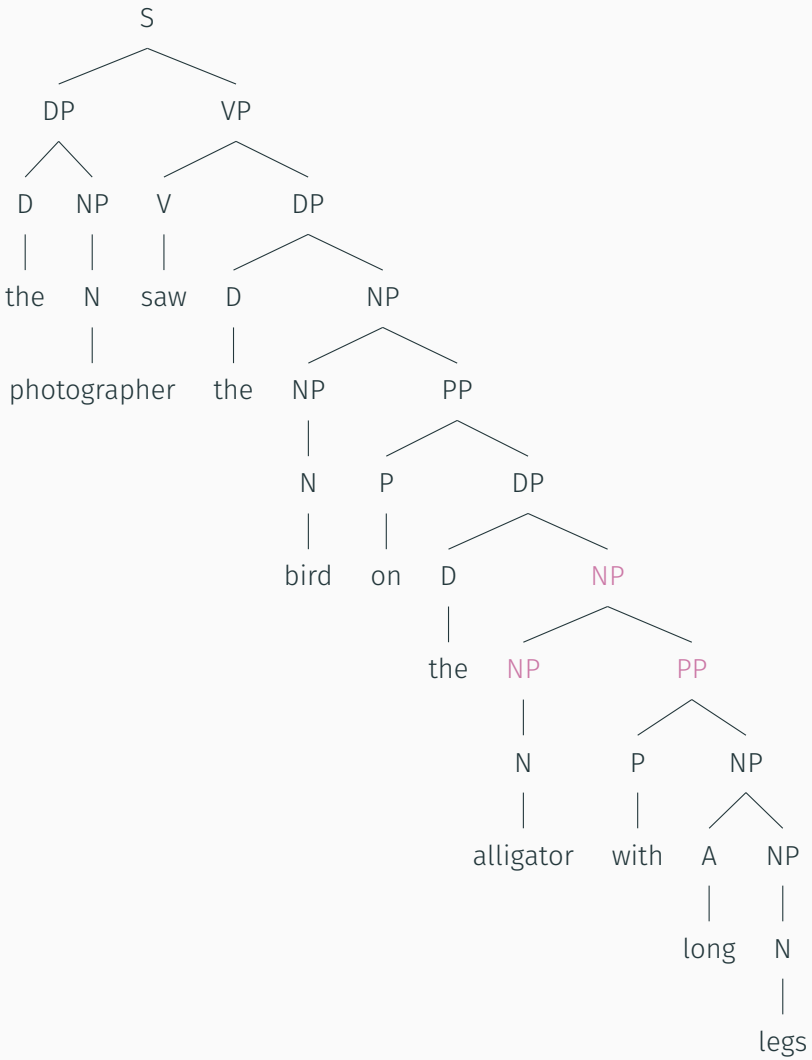
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- Reading:

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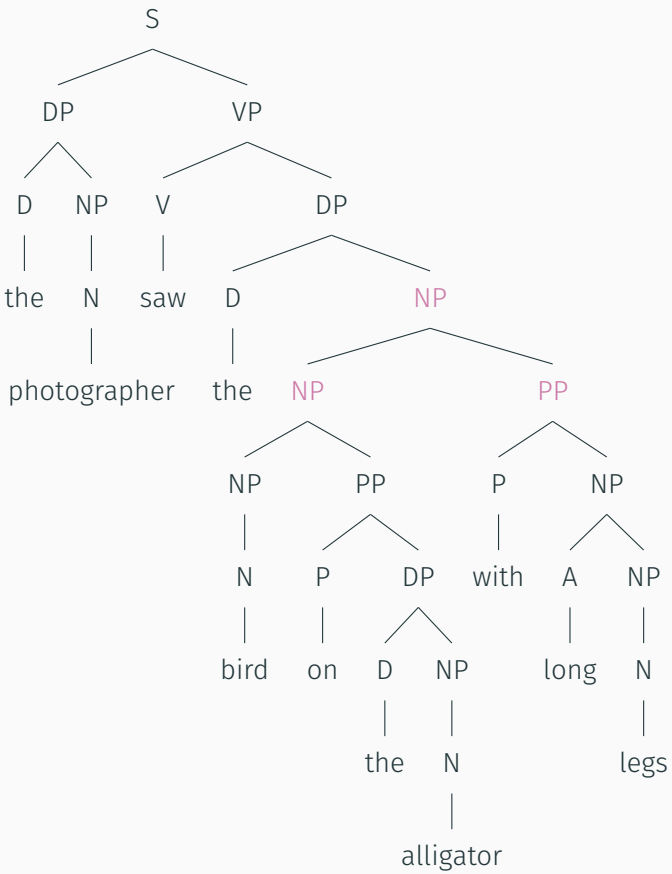
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- Reading: The photographer saw a bird that is on an alligator and the alligator has long legs.

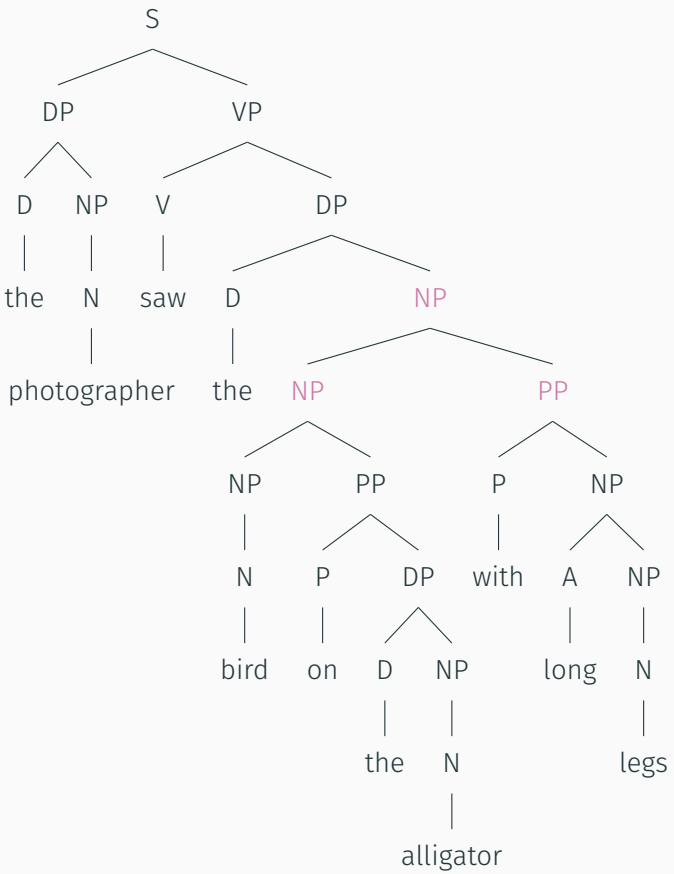
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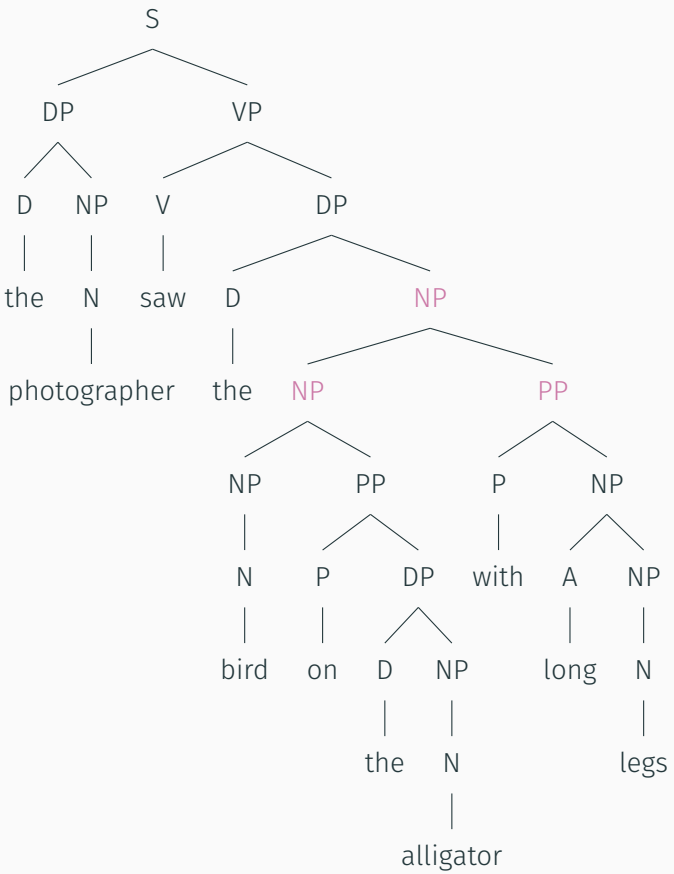
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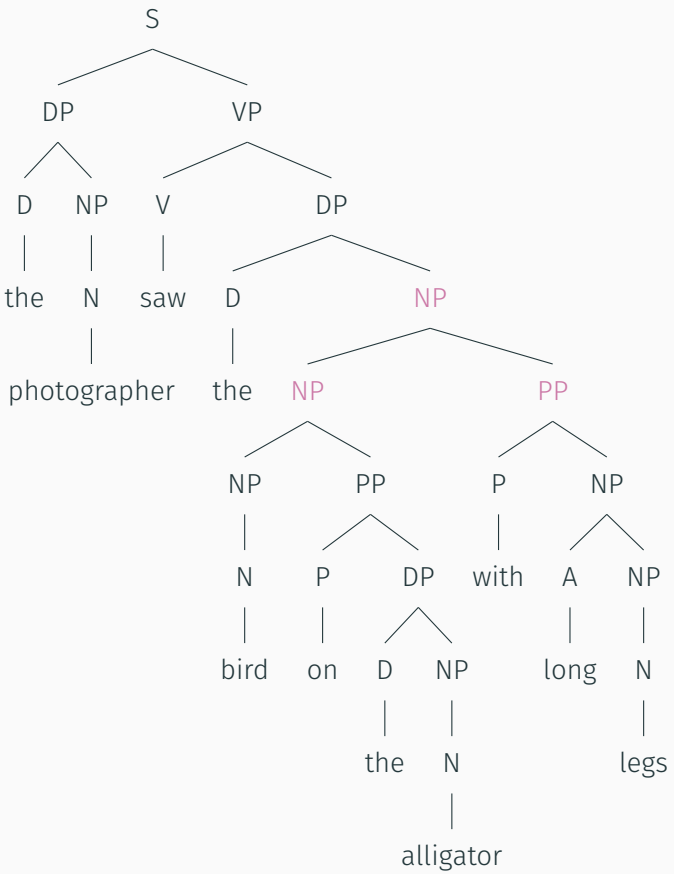
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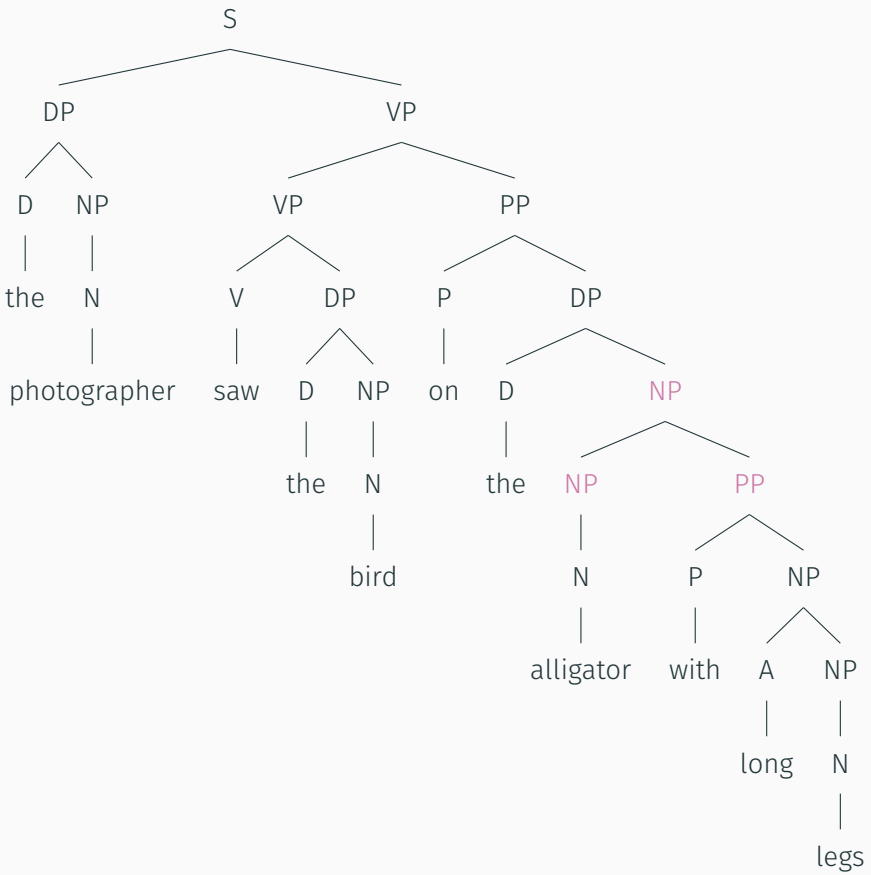
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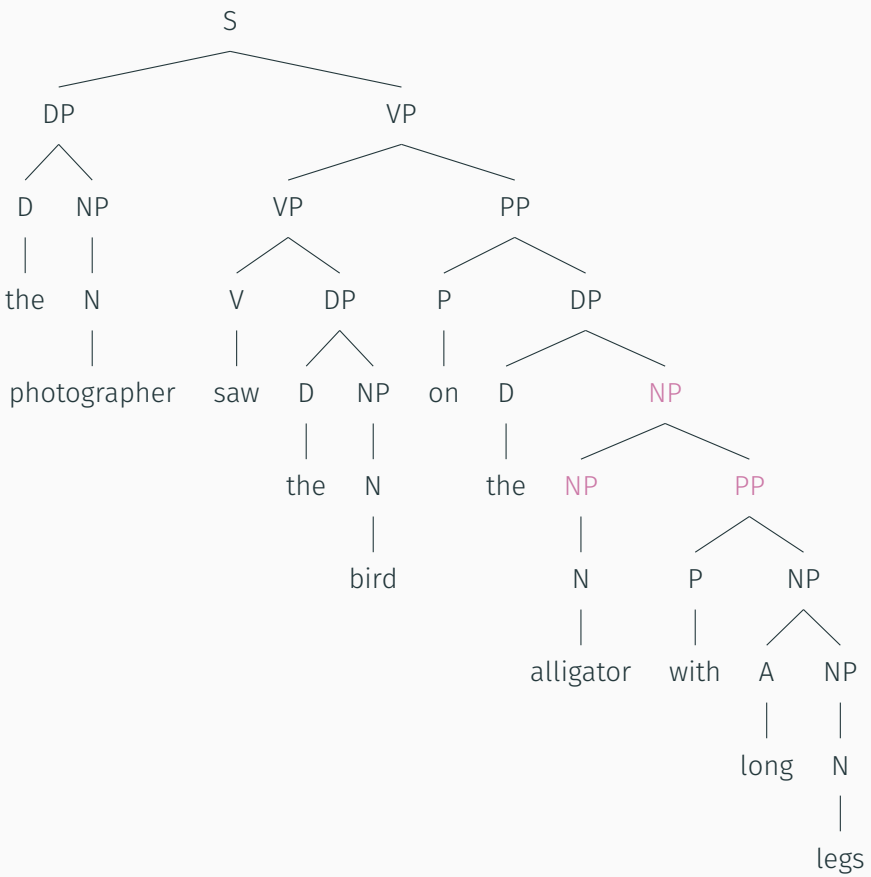
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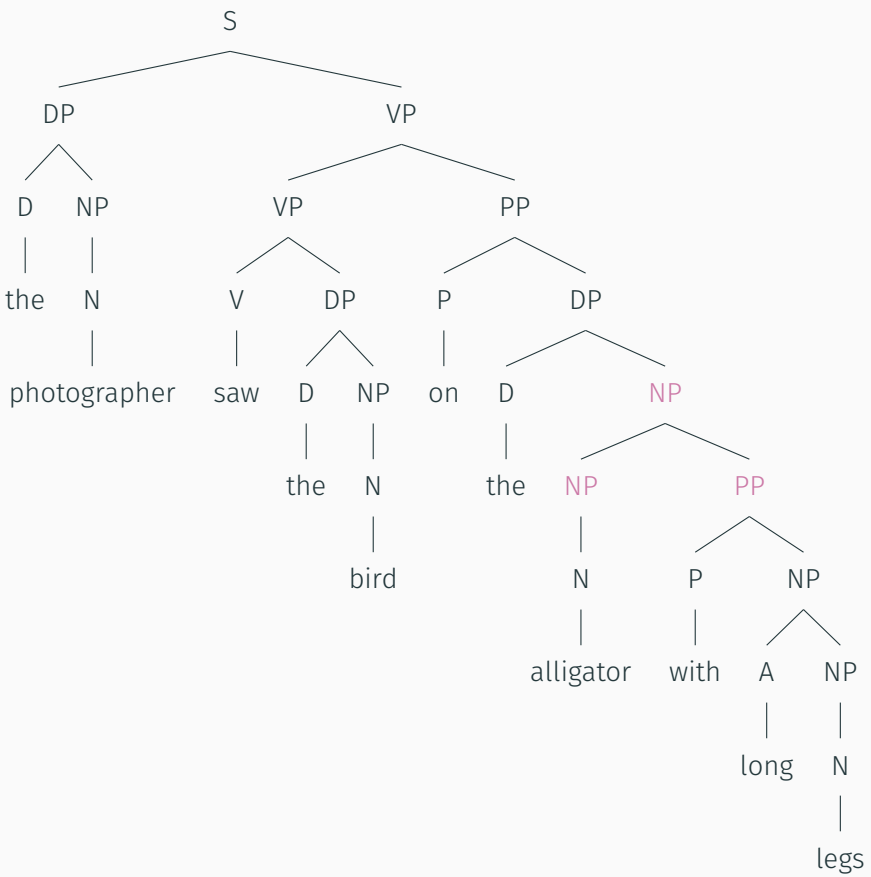
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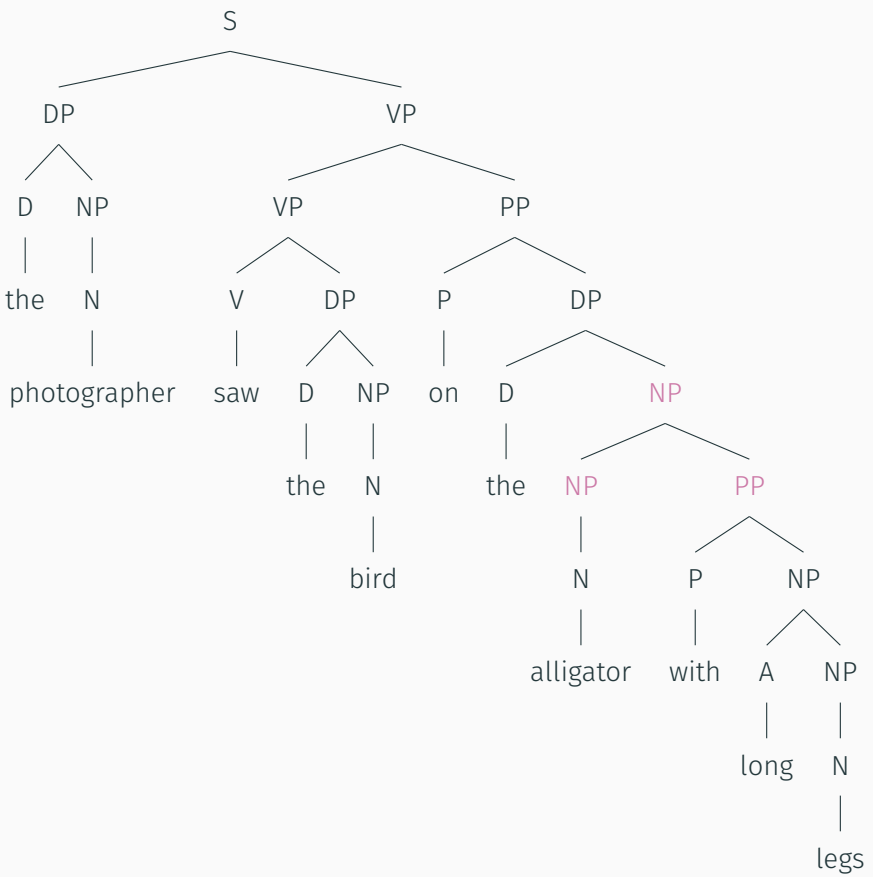
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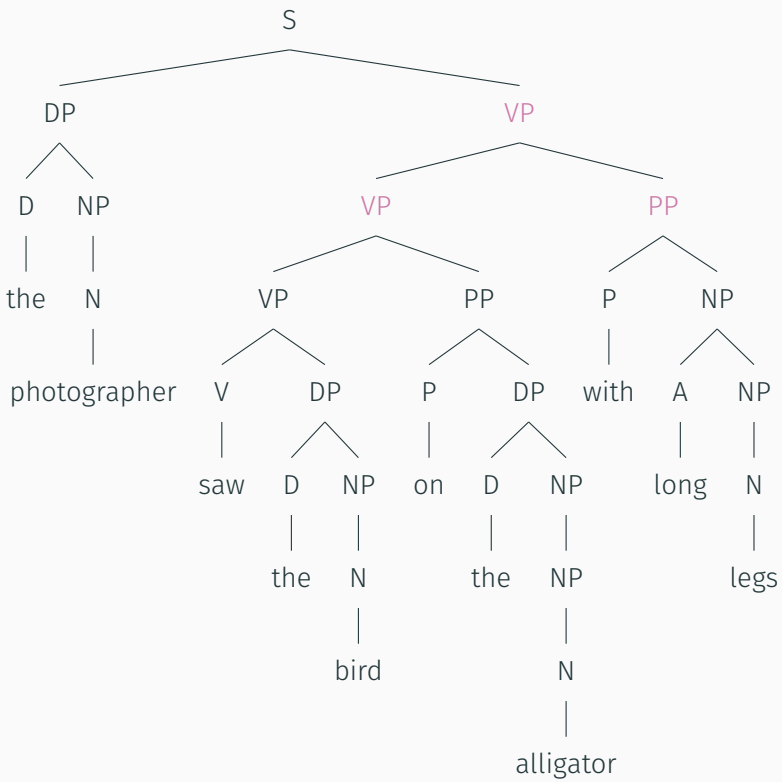
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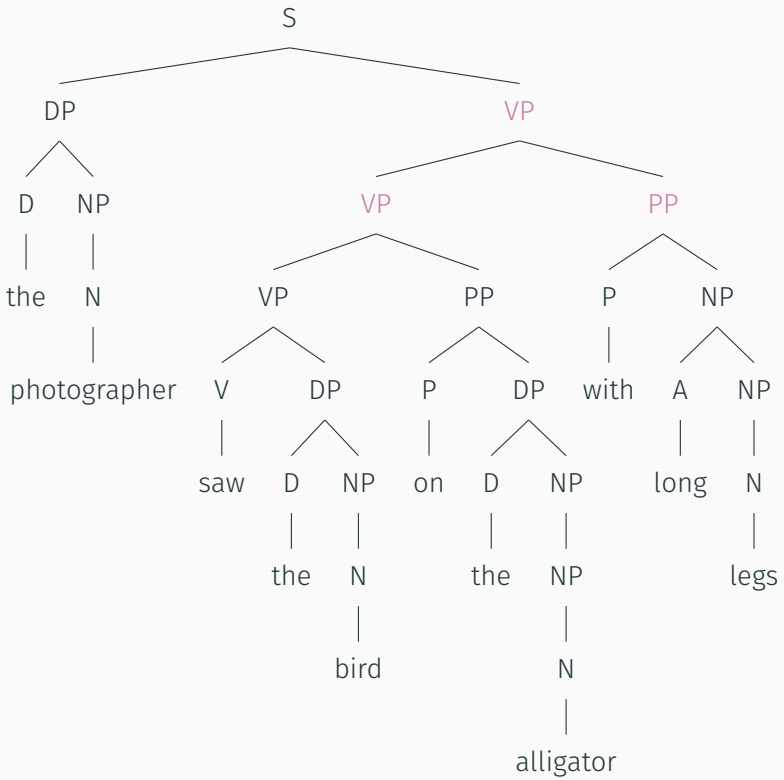
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Structural ambiguity 2': double PP-attachment

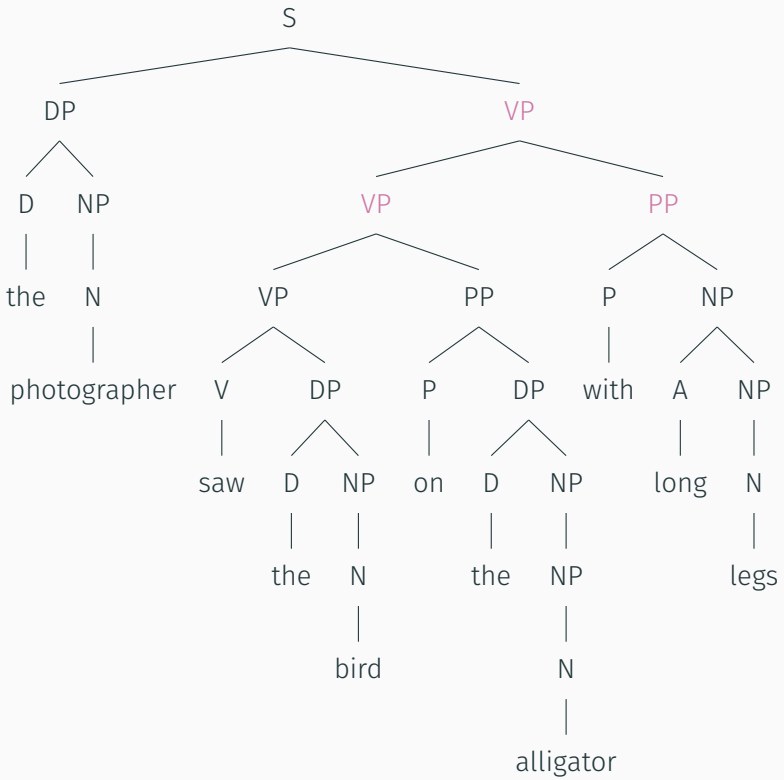
(1) The photographer saw the bird on the alligator with long legs.



- We modify the photographer-on-alligator tree by applying an extra VP $\rightarrow$ VP PP rule to modify *saw* with *with long legs*.

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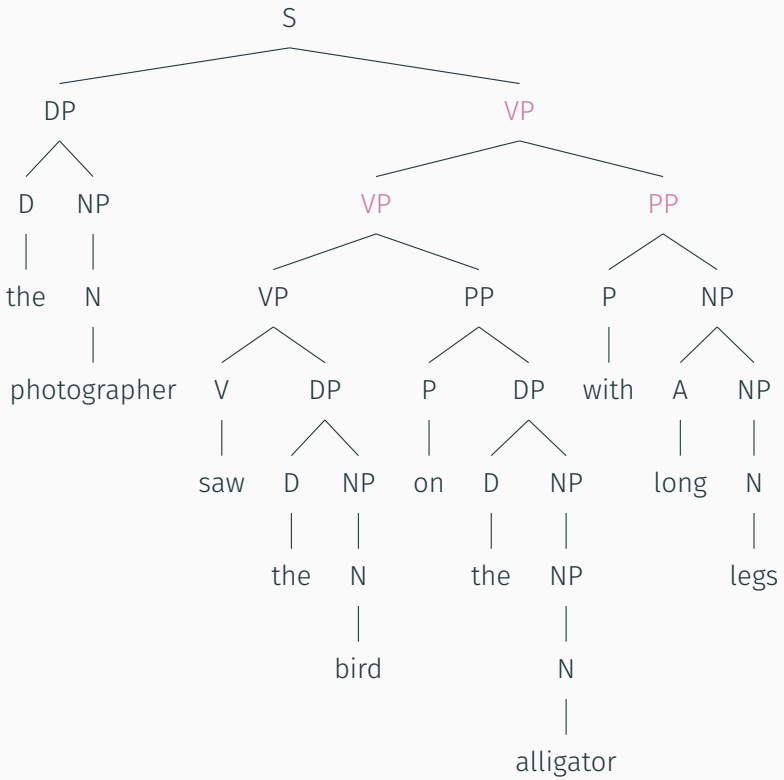
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- Reading:

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(1) The photographer saw the bird on the alligator with long legs.



- We modify the photographer-on-alligator tree by applying an extra VP $\rightarrow$ VP PP rule to modify *saw* with *with long legs*.
- Reading: The photographer saw a bird and the photographer did so on an alligator "using" long legs. Possible but semantically undesired![1]

[1] Why is it a semantic issue? Replace *long legs* by *powerful binoculars*... the reading now makes sense!

Context-free grammar and structural ambiguities recap

- Context-free grammars primarily generate strings by **calling production rules** (i.e. branching), until all the outputs are terminals.
- But interestingly, keeping track of which production rules were used to generate the final string (and what was their exact timing), sometimes allows us to **derive distinct syntax trees for the same string**–cashing out semantic ambiguities.
- In that sense, context-free grammars have a **strong generative capacity** when it comes to adjective and PP attachment ambiguities.
- The next (Bonus) section shows that context-free grammar also capture center-embedding constructions, that regexes could not match.

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Bonus: center-embedding

Center-embedding reminder

(2) a. The mouse is scared.

b. The mouse [that the cat chases] is scared.

c. The mouse [that the cat [that the dog barks at] chases] is scared.

d. The mouse [that the cat [that the dog [that the postman avoids] barks at] chases] is scared.

Center-embedding reminder

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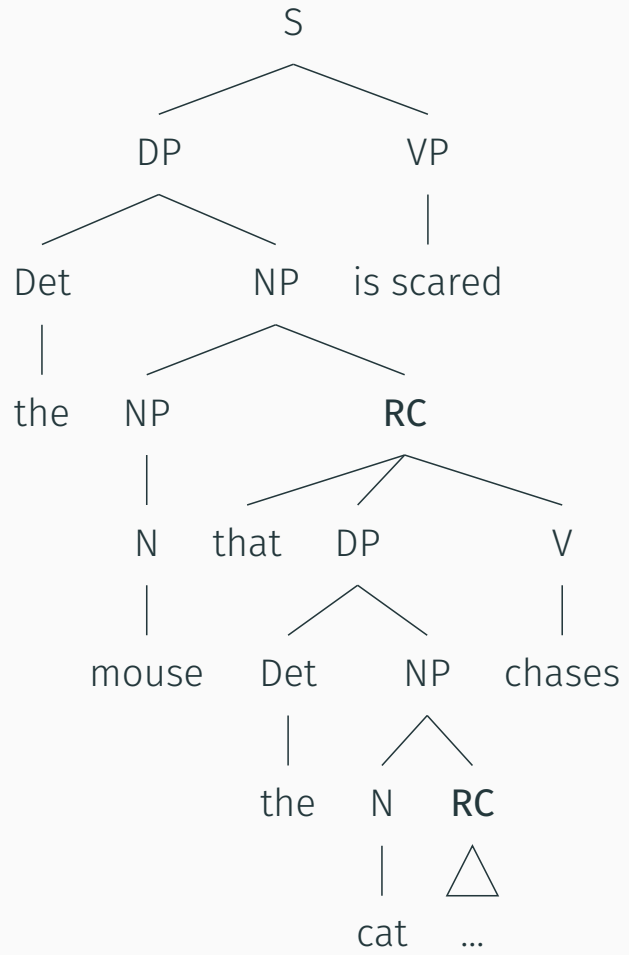
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Center-embedding: tree

- Recall the center-embedding pattern that regexes and automata could not capture: NP $\underbrace{(\text{that NP})^k(\text{V})^k}_{\text{nested relative clauses}}$ VP, **for any** $k \geq 0$.

- We treat relative clauses (**RCs**) a bit like **complex adjectives**: they may attach to NPs, below the determiner.
- Trick: RCs themselves contain NPs, which themselves may contain other RCs! This makes RCs **recursive**: RCs may introduce new, nested RCs.
- Ultimately, this enables center-embedding.

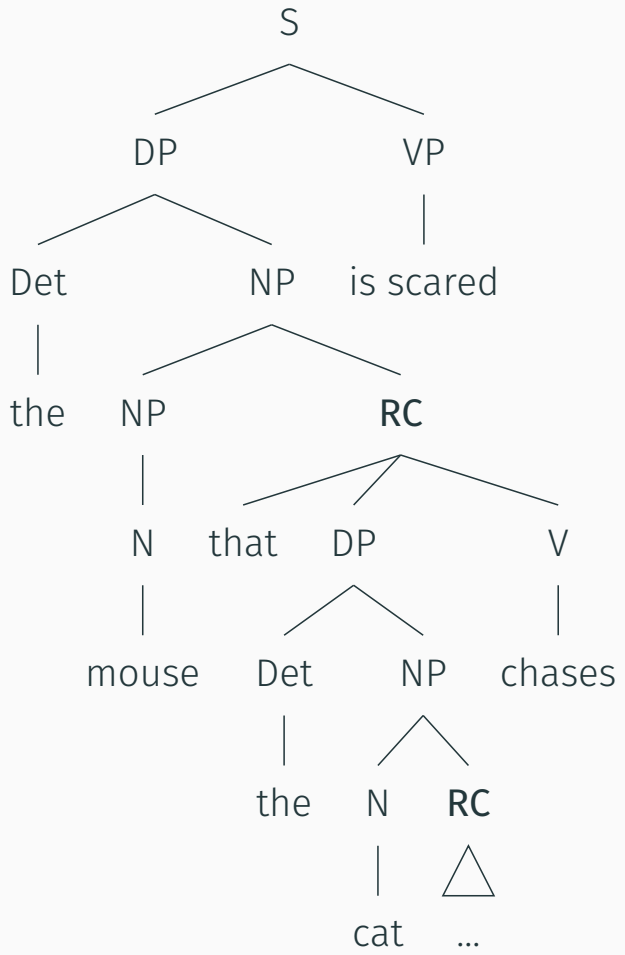


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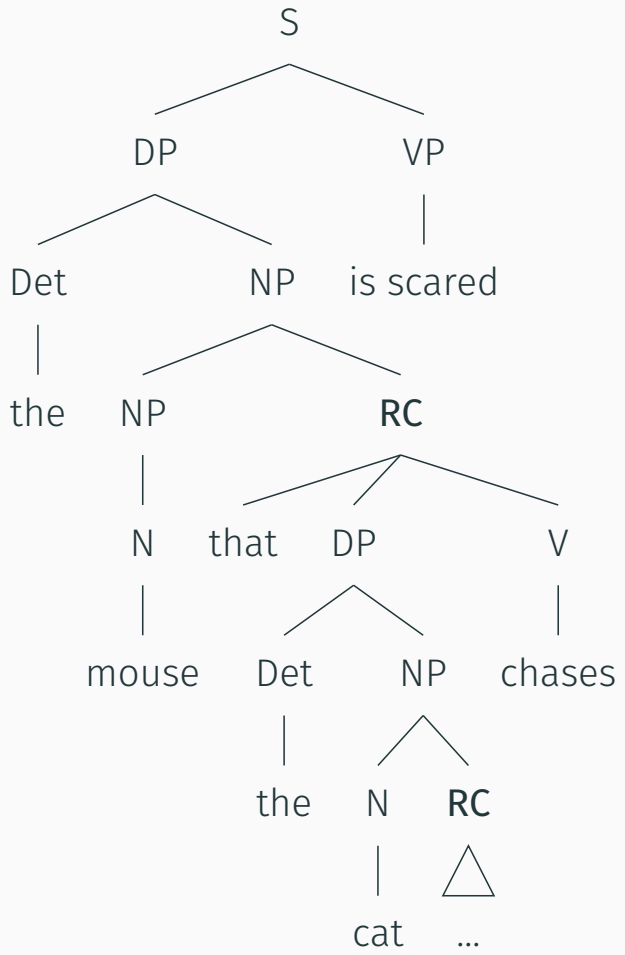
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Center-embedding: critical production rules

- Focusing on the nonterminal production rules:

$S \rightarrow DP\ VP$

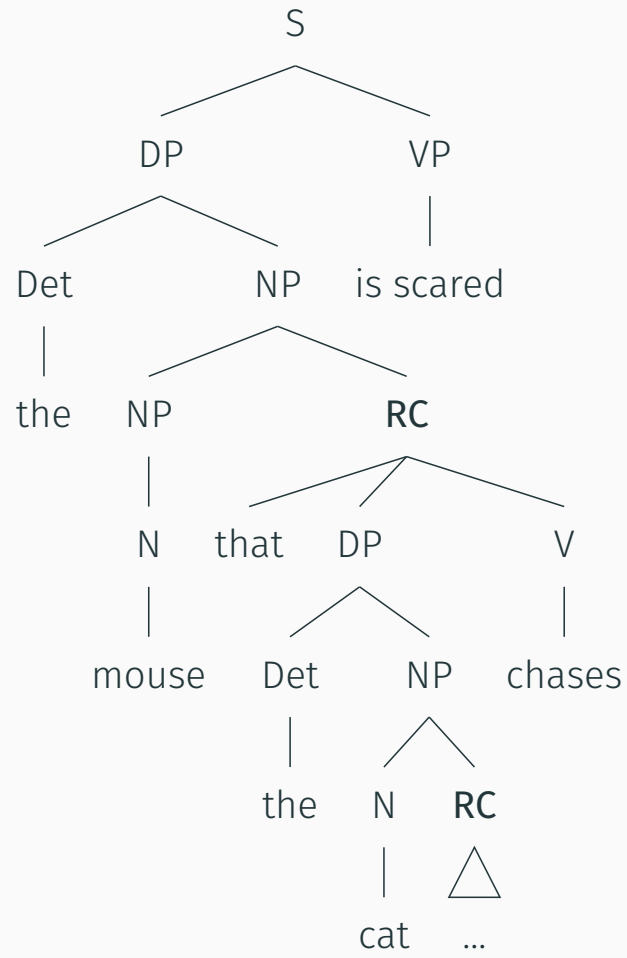
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- NPs introduce RCs or simply yield a N (crucial base case, for the derivation to stop at some point!)
- RCs in turn introduce DPs, which reintroduce NPs, which may reintroduce RCs!



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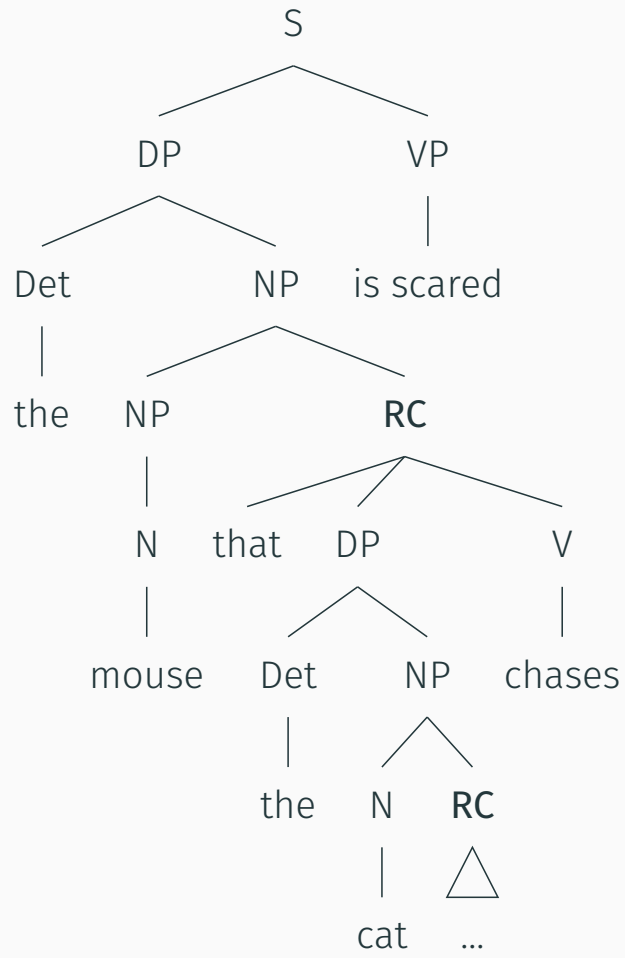
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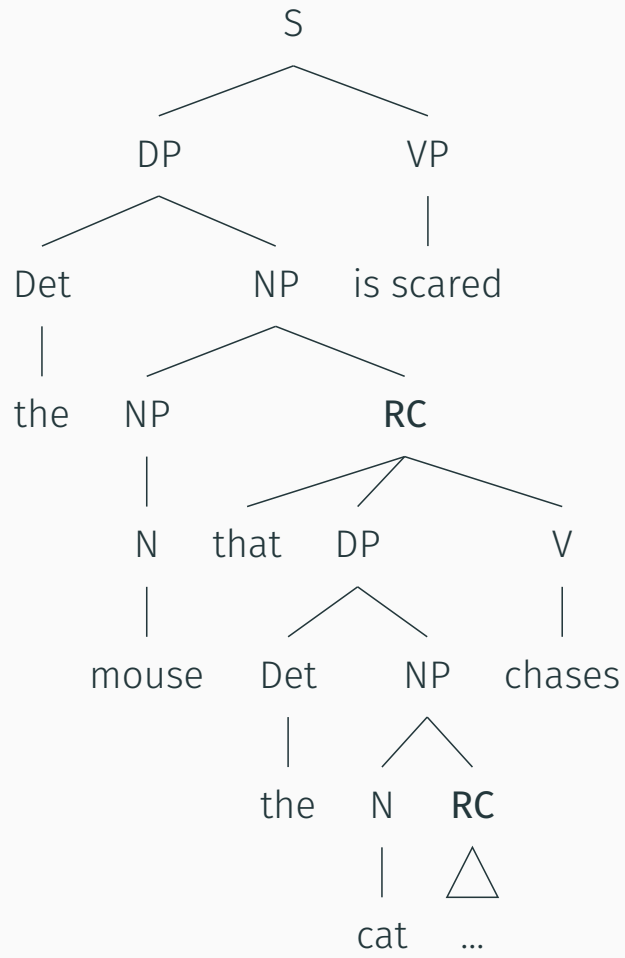
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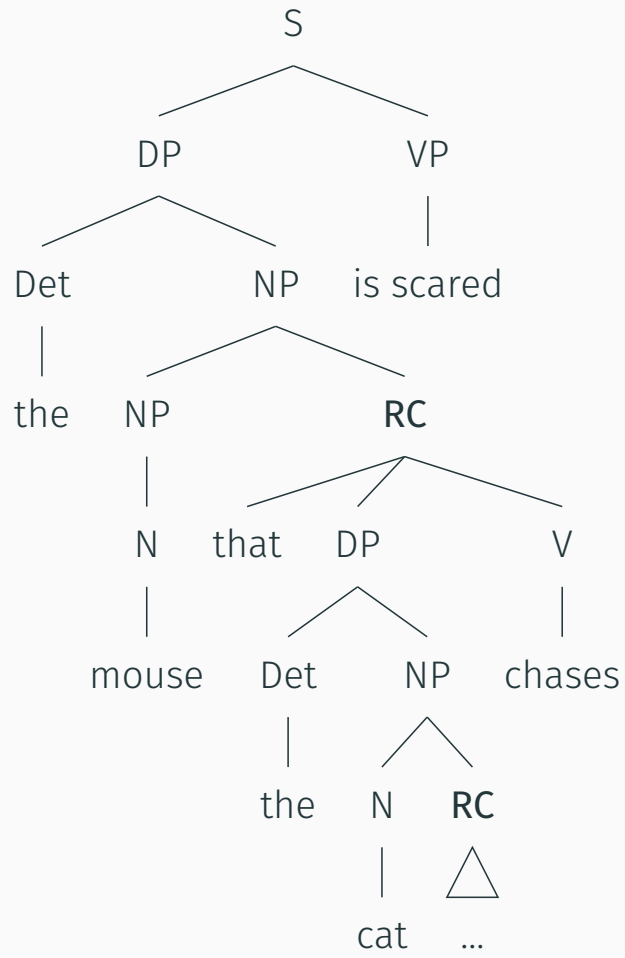
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Challenge: cross-serial dependencies

(3) Swiss German:

…mer em Hans es huus hälfed aastriiche.
…we Hans.DAT the house.ACC help paint.

'… we helped Hans paint the house.'

- In Swiss German, **predicates and their argument can be all jumbled up**! Nevertheless, predicates overtly **case-mark** their arguments, with Dative (in the case of *help*) and Accusative (in the case of *paint*).
- If we wanted to design a context-free grammar keeping track of predicate-argument dependencies (and of the resulting case-marking patterns), we'd struggle in the case of Swiss German, precisely **because the dependencies cross**.
- This pattern is quite rare cross-linguistically; dependencies tend to be nested. But it suggests that natural languages are in fact **more expressive than context-free grammars**.

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Lab

https://colab.research.google.com/drive/1Hh5Xz3awu5_EDNQPXdql-sTnZkPAfuizr?usp=sharing